

Supplementary Materials

Dataset Construction Documentation

Submitted in support of the main manuscript

Data retrieved from NCBI Virus SARS-CoV-2 Data Hub on 2024-03-15.

Lineage assignment by Pangolin with UShER (versions 4.3.4/1.37.1/v0.1.12/0.3.19/0.6.6).

Supplementary Table S.Y Per-Class Preprocessing and Sequence Quality Summary

This table reports, for each class: the raw pool size returned by the NCBI Virus query (Host = *Homo sapiens*; Nucleotide Completeness = complete, except Class 8); the number of sequences excluded by the >5% ambiguous-base (N) threshold; the sequence length range; the median N-content; whether random subsampling was applied; and the final sequence count used in the study. Subsampling was performed without replacement at a fixed random seed of 42 (Python `pandas DataFrame.sample`).

Supplementary Table S.Y. Per-class preprocessing and sequence quality summary.

Class	Lineage	Raw Pool	Excl. (>5% N)	Length Range (nt)	Median N%	Subsamp.?	Final N	Notes
0	B.1.1.519	7,872	0	29,576–29,918	0.0000	Yes	1,000	Complete only
1	B.1.1.529 (Omicron)	2,251	406	29,379–29,912	0.0000	Yes	1,000	Complete only
2	B.1.1.7 (Alpha)	278,603	141 [†]	29,579–29,903	0.4049 [†]	Yes	1,000	Complete only; stats from 2,000-seq random subset
3	B.1.351 (Beta)	1,358	79	29,636–29,903	0.1746	No	403	Complete only; all retained
4	B.1.427	5,884	0	29,552–29,904	0.0000	Yes	1,000	Complete only
5	B.1.429	12,802	0	29,430–29,912	0.0000	Yes	1,000	Complete only
6	B.1.525	1,037	41	29,503–29,903	0.0201	No	431	Complete only; all retained
7	B.1.526	21,801	1	29,556–29,903	0.0000	Yes	1,000	Complete only
8	B.1.621	4,805	417	27,919–29,909	0.0034	Yes	1,000	Complete + partial*; 975 partial, 25 complete
9	C.37 (Lambda)	1,255	20	29,310–29,903	0.0134	No	400	Complete only; all retained
10	P.1 (Gamma)	9,937	62	29,409–29,912	0.0000	Yes	1,000	Complete only
Total		342,897	1,167	—	—	—	9,234	

*The full pool was used for final subsampling.

N-content threshold: sequences with >5% ambiguous bases (N) were discarded; retained sequences ($\leq 5\%$ N) had ambiguous positions substituted with the dominant nucleotide at that position (per-lineage majority-consensus).

Length threshold: sequences <29,000 nt excluded (reference genome NC_045512.2 = 29,903 nt).

Subsampling: random without replacement, `random_state` = 42.

Supplementary Table S.Z: GenBank Accession IDs

This table provides the complete list of GenBank accession IDs (with version numbers) for all 9,234 SARS-CoV-2 sequences used in this study, organised by class label. Columns: Class (integer label, 0–10); Accession (GenBank accession with version, e.g., 0Z286432.1); Pango Lineage (as assigned by Pangolin with UShER, versions 4.3.4/1.37.1/v0.1.12/0.3.19/0.6.6); Collection Date; Nucleotide Completeness (as reported by NCBI Virus); Sequence Length (nt); PangoVersions (tool version string).

Class summary:

Class	Lineage	N	Collection Date Range	Completeness
0	B.1.1.519	1,000	2020 – 2022-06-19	complete
1	B.1.1.529 (Omicron)	1,000	2020 – 2023-02-09	complete
2	B.1.1.7 (Alpha)	1,000	2020 – 2021-08-24	complete
3	B.1.351 (Beta)	403	2020 – 2022-11	complete
4	B.1.427	1,000	2020-11-10 – 2022-06-12	complete
5	B.1.429	1,000	2020-10-14 – 2022-04-24	complete
6	B.1.525	431	2020 – 2022-07	complete
7	B.1.526	1,000	2020 – 2021-08-09	complete
8	B.1.621	1,000	2020-12-21 – 2021-11-19	complete
9	C.37 (Lambda)	400	2021-01-29 – 2021-11-20	complete
10	P.1 (Gamma)	1,000	2020 – 2022-06-25	complete
Total		9,234	2020 – 2023-06-07	

Sample accession entries (first 3 per class):

Supplementary Table S.Z GenBank accession IDs (first 3 entries per class shown; complete list in `final_accession_list.FINAL.csv`).

Class	Accession	Pangolin	Collection Date	Completeness	Length (nt)
0	MZ471646.1	B.1.1.519	2021-03-23	complete	29,782
0	OK435740.1	B.1.1.519	2021-02-07	complete	29,823
0	MZ202426.1	B.1.1.519	2021-04-15	complete	29,836
1	OM245852.1	B.1.1.529	2021-12-18	complete	29,756
1	ON313036.1	B.1.1.529	2022-02-02	complete	29,775
1	OP051818.1	B.1.1.529	2022-06-25	complete	29,809
2	OX561489.1	B.1.1.7	2021-04-14	complete	29,828
2	MZ009435.1	B.1.1.7	2021-04-01	complete	29,805

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Class	Accession	Pangolin	Collection Date	Completeness	Length (nt)
2	OX660204.1	B.1.1.7	2021-03-31	complete	29,884
3	OX446986.1	B.1.351	2021-05-18	complete	29,896
3	MZ970738.1	B.1.351	2021-03-31	complete	29,764
3	OZ289872.1	B.1.351	2021-02-25	complete	29,885
4	PQ967276.1	B.1.427	2021-05-11	complete	29,771
4	MW808430.1	B.1.427	2021-03-07	complete	29,872
4	OP143274.1	B.1.427	2021-01-05	complete	29,825
5	MZ185912.1	B.1.429	2021-04-16	complete	29,894
5	MZ500442.1	B.1.429	2021-03-12	complete	29,824
5	MZ635657.1	B.1.429	2021-01-26	complete	29,903
6	OX613787.1	B.1.525	2020	complete	29,903
6	ON030839.1	B.1.525	2021-04-23	complete	29,773
6	MZ362445.1	B.1.525	2021-05-18	complete	29,693
7	MZ153577.1	B.1.526	2021-04-20	complete	29,818
7	MW891313.1	B.1.526	2021-03-20	complete	29,817
7	MW966199.1	B.1.526	2021-03-24	complete	29,891
8	OM433704.1	B.1.621	2021-07-08	complete	29,885
8	OY313236.1	B.1.621	2021-07-12	complete	29,823
8	OL533323.1	B.1.621	2021-07-24	complete	29,902
9	OZ064465.1	C.37	2021-08-04	complete	29,768
9	MZ076366.1	C.37	2021-04-08	complete	29,843
9	MZ579108.1	C.37	2021-04-19	complete	29,894
10	MZ283260.1	P.1	2021-04-21	complete	29,841
10	MZ331233.1	P.1	2021-05-03	complete	29,844
10	MZ308082.1	P.1	2021-05-10	complete	29,709
Total (full list in CSV file)					9,234

Note: The complete 9,234-row accession list is provided in the accompanying file `final_accession_list_FINAL.csv` with columns: Class, Accession, Pangolin, Collection_Date, Nuc_Completeness, Host, Length, PangoVersions. Sequences can be retrieved directly from NCBI GenBank at <https://www.ncbi.nlm.nih.gov/nuccore/> using the accession IDs provided. All PangoVersions values are uniformly 4.3.4/1.37.1/v0.1.12/0.3.19/0.6.6 across all 9,234 records.